

一. 狭义相对论

1. 4维时空

模型设定: 事件发生于-瞬间, 全部事件的集合 $\rightarrow$ 时空

event

质点 $\rightarrow$ 粒子, 位置在某个具体的点.

particle

宏观近似

$\downarrow$

$$\lim \Delta t \rightarrow 0$$

$$\lim \Delta x \rightarrow 0$$

(观测者的集合) 参考系 $\rightarrow$ 时空(或-子集)的任一点仅有1个观测者的世界线经过

reference frame

观测者有标准钟 $\rightarrow$ 观测者的固有时

standard clock

proper time

闵氏时空: $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \rightarrow$ 对光 $|x| = ct$, $ds^2 = 0$
对洛伦兹变换 $L = \begin{bmatrix} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{bmatrix}$, 对3维 $\rightarrow$ 经典旋转

$$\text{速率 } u := \frac{\sqrt{dx^2 + dy^2 + dz^2}}{dt}$$

光: $u = c$, $ds^2 = 0$

质点: $u < c$, $ds^2 > 0$

光速不变原理

各观测者无特殊性] 狭义相对性原理

时空的微分几何语言:

观测者对自己: ∂_a (普通导数) $\rightarrow \partial_a \eta_{bc} = 0$

速率 $u = 0 \rightarrow$ 世界线上 $\left. \begin{array}{c} t \\ | \\ x \end{array} \right|_x \rightarrow \partial_a (\partial/\partial t)^b = 0 \rightarrow (\partial/\partial t)^a \partial_a (\partial/\partial t)^b = 0$

洛伦兹变换: 保度映射: $\phi: M \rightarrow M$. $\phi^* \eta_{ab} = \eta_{ab}$. $ds^2 = ds'^2$

2. 相对论四维量

4速
4-velocity

$$u^a := (\partial/\partial\tau)^a$$

$$u^a u^b g_{ab} = -1 \quad \text{pf: } u^a \text{ 为线元的导 (类时曲线)}$$

相对观察者: 用相交世界线观测 W_p 为 u^a 的法空间 ($w_a u^a = 0$). W_p 为空间

在观测者 G 上, 法空间的诱导度规 $h_{ab} \equiv g_{ab} + Z_a Z_b$ ($n_a n^a = -1$) $\rightarrow h^a_b v^b = v^a$ 为投影空间度

$$u^a = (\frac{\partial}{\partial\tau})^a = (\frac{\partial}{\partial t})^a \frac{dt}{d\tau} + (\frac{\partial}{\partial x^i})^a \frac{dx^i}{d\tau} \rightarrow \text{速度 } u^a := (\frac{\partial}{\partial x^i})^a \cdot \frac{dx^i}{d\tau} = (\frac{\partial}{\partial x^i})^a \cdot \frac{\frac{dx^i}{dt}}{\frac{dt}{d\tau}} = h^a_b u^b \cdot \gamma$$

$\frac{dt}{d\tau} = \gamma$ 或 $\gamma = -u^a Z_a = \frac{1}{\sqrt{1-v^2}}$

分解: $u^a = \gamma(Z^a + u^a)$, $\gamma = -Z^a u_a$, $\gamma u^a = u^a - \gamma Z^a$

4动量

$$P^a := m u^a \rightarrow P^a = E \cdot Z^a + p^a \quad E = -P^a \cdot Z_a \quad \text{4动量守恒} \quad P^a = \bar{P}^a$$

$$P^a P_a = -m^2 = -E^2 + p^2 \rightarrow E^2 = m^2 + p^2 \quad (c=1)$$

4加速度

$$A^a := u^b \partial_b u^a \quad \text{在世界线上定义 } \frac{du^a}{d\tau}$$

1) 是绝对的 2) $A^a = 0 \leftrightarrow u^b \partial_b u^a = 0 \leftrightarrow$ 惯性运动

$$A^a u_a = 0 \quad \text{pf: } A^a u_a = u^b u_a \partial_b u^a = \frac{1}{2} u^b \partial_b (u_a u^a) = 0$$

分解: $\equiv$ 加速度: $a^a := \frac{d^2 x^i(t)}{dt^2} (\frac{\partial}{\partial x^i})^a$, $A^a = A^a (dx)_a^a = (dx)_a^a u^b \partial_b u^a = u^b \partial_b \frac{u^a}{\gamma} = \frac{du^a}{d\tau} = \gamma \frac{du^a}{dt}$

4力

$$F^a := u^b \partial_b P^a \quad \frac{dm}{d\tau} = 0 \text{ 时, } F^a = m A^a$$

$$F^0 = \gamma \frac{dE}{dt}$$

$$F^i = \gamma f^i$$

$$\uparrow \text{功率, 注意 } \vec{f} \cdot \vec{u} = \frac{dE}{dt} - \gamma^2 c^2 \left(\frac{dm}{dt} \right) = \frac{dE}{dt}$$

$$A^0 = \gamma \frac{d\gamma}{dt} = \gamma^3 \vec{u} \cdot \vec{a}$$

$$A^i = \gamma \frac{d\gamma u^i}{dt} = \gamma^3 a^i + u^i \gamma \frac{d\gamma}{dt} = \gamma^3 a^i + \gamma^3 (\vec{u} \cdot \vec{a}) u^i$$

$$A=0 \rightarrow A^0=0; A^i=0 \Rightarrow a^i=0$$

能量动量张量

$$T_{ab} = T_{ba}$$

↓

4动量密度

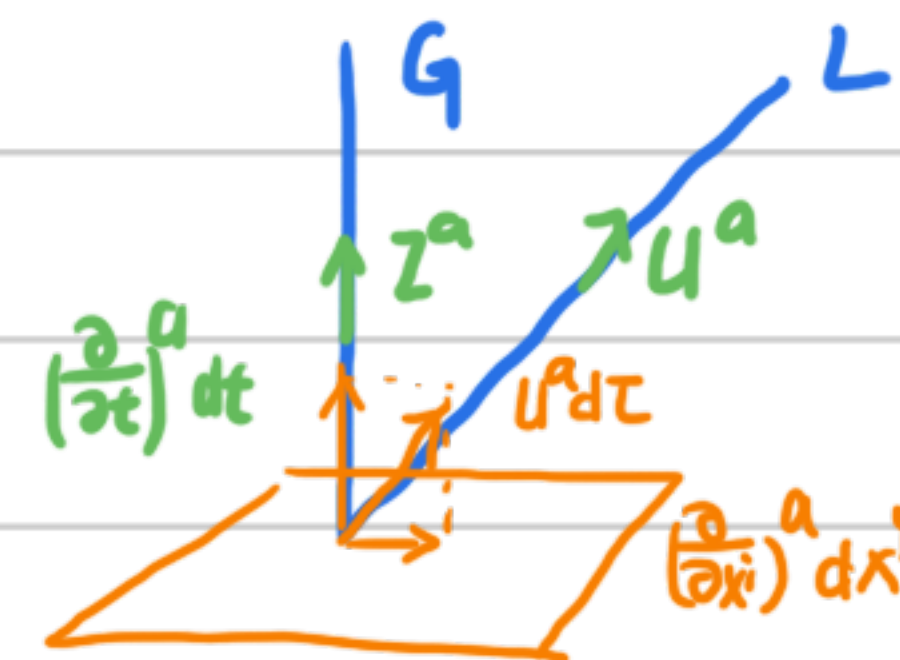
$$W^a := -T^{ab} u_b = \mu u^a + \underline{w^a}_{w^i(e_i)^a}$$

对观测者

$$\mu \equiv T_{ab} u^a u^b = T_{00} \text{ 能量密度}$$

$$w_i \equiv -T_{ab} u^a (e_i)^b = -T_{0i} \text{ 能流密度 (动量流)}$$

$$T_{ij} = T_{ab} (e_i)^a (e_j)^b \text{ 应力张量}$$



$$\partial_a T^{ab} = 0 \rightarrow \text{守恒量都守恒}$$

pf: 能守: $\partial_a W^a = \partial_a (-T^{ab} Z_b) = -\partial_a T^{ab} \cdot Z_b - T^{ab} \underbrace{\partial_a Z_b}_{\partial_a (dt)_b = 0}$
(高斯定理可证)

理想流体: $m \rightarrow \rho \rightarrow T_{ab} = \mu \cdot u_a u_b + p(\eta_{ab} + u_a u_b)$

静止观者: $(e_0)^a = u^a|_{P_x}$

$$T_{00} = \mu + p(-1+1) = \mu$$

$$T_{ij} = p(\delta_{ij}), T_{i0} = 0$$

牛顿流体: $\partial_t \mu + \nabla \cdot (\mu \cdot \vec{u}) = 0$ (质量守恒)

相对论: 无外力 $\partial^a T_{ab} = 0 = u_b u_a \partial^a (\mu + p) + (\mu + p) \partial^a u_a u_b + \partial_b p$

投影时间 $\rightarrow 0 = u^b \partial^a T_{ab} = -u_a \partial^a (\mu + p) - (\mu + p) \partial^a u_a + u^b \partial_b p + 0$

$$= -u_a \partial^a (\mu) - (\mu + p) \partial^a u_a$$

投影空间 $\rightarrow h_c^b = \eta_c^b + u_c u^b \rightarrow h_c^b \partial^a T_{ab} = 0 = (\mu + p) u^a \partial_a u_c + \partial_c p + u_c u^b \partial_b p$

非相对论: $\gamma \rightarrow 1, u^a \approx (u^0)^a + u^a$
 $\mu \gg p$
 $u \ll 1$

$$\begin{cases} 0 = \partial_t \mu + \mu \nabla \cdot (\mu \cdot \vec{u}) \\ 0 = \mu \left(\frac{\partial u_i}{\partial t} + u^a \partial_a u_i \right) + \partial_i p + u_i \partial_t p + u_i u^j \partial_{ji} p \end{cases}$$

$< \partial_i p, \partial_{ji} p$

电动力学 (见朗道《场论》笔记)